

DisSModel: A Python Framework for Spatially Explicit Dynamic Modeling

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Summary

DisSModel (Discrete Spatial Modeling) is a modular Python framework for spatially explicit dynamic modeling, targeting the complexities of Land Use and Land Cover Change (LUCC). It translates the modeling paradigms of the TerraME framework ([Carneiro et al., 2013](#)) into the Python ecosystem, enabling researchers to simulate complex socio-environmental systems — forest fires, epidemiological spreads, coastal dynamics — by coupling a time-stepped clock with the spatial data structures of GeoPandas ([Jordahl et al., 2021](#)).

The framework provides a **dual-substrate architecture**: a vector substrate backed by GeoDataFrame for spatial expressiveness, and a raster substrate backed by NumPy 2D arrays for high-performance vectorised computation. DisSModel is available through the DisSModel GitHub organisation and on PyPI.

Statement of Need

Python has become the lingua franca for geospatial data science, supported by libraries such as GeoPandas and PySAL — but these tools target static analysis. Dynamic spatial modeling, simulating how landscapes evolve over time, has historically required specialised platforms. In Brazil, TerraME ([Carneiro et al., 2013](#)) and Dinamica EGO are the most widely adopted general-purpose frameworks, while institutions elsewhere rely on narrower allocation models such as CLUE and CLUE-S ([Veldkamp & Fresco, 1996](#)). This fragmentation leaves researchers choosing between a Lua-based toolchain and single-purpose implementations with no shared contract.

While TerraME is conceptually robust, its reliance on Lua — a language with far smaller adoption in data science than Python — creates a barrier for data scientists, and the framework has seen no new release since August 2020. General-purpose simulation libraries, meanwhile, lack native synchronisation between a time-stepped clock and the geographical state of a GeoDataFrame. DisSModel fills this gap with a lightweight scheduler coupled directly to vector and raster spatial state, providing a Pythonic, actively maintained implementation of the TerraME paradigm — and, through its satellite packages, of allocation models like CLUE and CLUE-S.

Reproducibility is a first-class concern: the executor module provides a standardised lifecycle — validate → load → run → save — capturing provenance metadata (input checksums, parameters, timing, output paths) in an ExperimentRecord generated automatically for every run, with no additional instrumentation by the modeller.

State of the Field

DisSModel occupies a niche between general-purpose agent-based modeling (ABM) libraries and specialised GIS simulation software:

Aspect	TerraME	Dinamica EGO	DisSModel
Language	Lua	Visual/Internal	Python
Simulation Engine	Discrete Event	Cellular Automata	Time-stepped scheduler
Spatial Structure	CellularSpace (Fixed)	Cellular Grid	GeoDataFrame + NumPy (Dual)
GIS Integration	TerraLib	Native Raster	GeoPandas / Rasterio
Extensibility	Script-based	Block-based	Class Inheritance
Reproducibility	Manual	Manual	Automated (ExperimentRecord)
Neighborhoods	GPM Support	Limited	libpysal weights (Queen, Rook, KNN, custom)

NetLogo and Mesa are excellent for ABM but require boilerplate to handle real-world spatial projections. DisSModel uses GeoPandas as its core engine, following the discrete spatial modeling approach of Santos Junior (2025).

Software Design

DisSModel is organised into five modules with strict separation of concerns, extensible through class inheritance. **Core** manages the simulation clock: the Environment orchestrates time progression, and spatial models auto-register at instantiation, receiving ticks through setup / pre_execute / execute / post_execute hooks. **Geo** provides the dual-substrate design: a vector substrate (SpatialModel, CellularAutomaton) on GeoDataFrame with libpysal neighbourhoods (Rey et al., 2021), and a raster substrate (RasterModel, RasterCellularAutomaton) on NumPy arrays with vectorised operations (shift2d, focal_sum, neighbor_contact) replacing cell-by-cell loops. **Executor** defines the ModelExecutor four-phase lifecycle (validate, load, run, save); subclasses self-register via __init_subclass__, and every run produces an ExperimentRecord with input checksum, parameters, timing, and output paths. **IO** provides a unified dataset abstraction (load_dataset / save_dataset) across GeoDataFrame, GeoTIFF, and Xarray/Zarr, with transparent s3:// resolution. **Visualization** integrates Matplotlib, Streamlit-compatible widgets, and RasterMap.

This extensibility has already produced independent domain packages: dissmode-ca (Costa & Santos Junior, 2025a) (Cellular Automata patterns), dissmode-sysdyn (Costa & Santos Junior, 2025b) (System Dynamics), and DisLUCC-Continuous (Costa et al., 2025), which implements LUCCME's continuous components — Demand, Potential, and Allocation (Veldkamp & Fresco, 1996; Verburg et al., 2004) — on both substrates and the same ModelExecutor contract, an explicit Python counterpart to TerraME/LUCCME.

Performance

The vector substrate offers spatial expressiveness; the raster substrate achieves high throughput via NumPy vectorisation. All benchmarks ran on an Intel Core i7-7700T @ 2.90GHz, 15 GB RAM (Ubuntu, Python 3.12.3, NumPy 2.4.6, GeoPandas 1.1.3); absolute timings vary by hardware, but the relative speedup is the result of interest.

Conway's Game of Life confirms mathematical equivalence across substrates with different throughput:

	Grid	Cells	Raster (ms/step)	Vector (ms/step)	Speedup
	10×10	100	0.12	74.81	639×
	50×50	2,500	0.19	1,707.76	8,809×

Grid	Cells	Raster (ms/step)	Vector (ms/step)	Speedup
100×100	10,000	0.41	7,069.14	17,394×
200×200	40,000	1.21	—	—
500×500	250,000	9.74	—	—
1,000×1,000	1,000,000	30.60	—	—

BR-MANGUE coastal dynamics. The foundation for coupled mangrove-flood modeling was established by Bezerra et al. (D. da S. Bezerra et al., 2013) and extended in D. da S. Bezerra et al. (2025), co-authored by Denilson da Silva Bezerra, the submitting author, and Felipe Martins Sousa — the same researchers responsible for the DisSModel reimplementation. The brmangue-dissmodel package (Costa & Bezerra, 2026) validates the raster implementation against TerraME over the Maranhão Island dataset (50,496 cells, 19 steps): land use and soil match exactly at every checkpoint (MAE 0, max error 0), and elevation on 97.3% of cells within 1 mm (MAE 0.00068 m) — match percentage being the appropriate metric for categorical outputs (Pontius Jr. et al., 2011). In this scenario the flood component triggers no land-use transition and the golden files confirm TerraME does the same, so the agreement above exercises mangrove migration; flooding is covered separately under the laboratory parameters. Reproducible via brmangue-dissmodel/src/brmangue/executors/validation_executor.py (end_time=19) against the committed golden CSVs in tests/fixtures/golden/, with tests/test_model_invariants.py and tests/test_transition_rules.py covering structural correctness.

Cross-substrate equivalence (60×60 synthetic grid, 3,600 cells, 10 steps) shows 100% match for land use, soil, and elevation under tolerance (MAE 0.000959 m, max error 0.024 m), with raster at 2.1 ms/step against 84.2 ms/step for vector (40.1× speedup); the residual elevation divergence is floating-point rounding, not algorithmic disagreement. Each run automatically produces an ExperimentRecord with timings, checksums, and artifact paths.

DisSLUCC-Continuous implements the continuous CLUE-like allocation algorithm (Veldkamp & Fresco, 1996); MAE is the appropriate metric for its fractional outputs (Pontius Jr. et al., 2011; Willmott & Matsuura, 2005). Over the Lab1 study area (6,574 cells, 6 steps), both substrates reproduce the TerraME/LuccME reference at MAE = 0.0036 (RMSE 0.0062, max error 0.027) in [0,1] scale. The residual is accounted for by the original model's own convergence tolerance: the LuccME script declares maxDifference = 1643 area units against a 2014 demand of 21,607 — a 7.6% band, within which the reference itself stops short of its declared demand. Consistently, the Pontius decomposition attributes 90% of the residual to quantity and 10% to allocation (Pontius Jr. & Millones, 2011). The raster substrate is 3.9× faster (44.0 ms/step vs 172.8 ms/step). Reproducible via disslucc-continuous/tests/test_benchmark_validation.py, with tests/test_benchmark_discriminance.py confirming that perturbing the regression coefficients breaks the tolerance criterion. End-to-end provenance from raw inputs to final metrics is addressed by the dissmodel-platform package.

Research Impact Statement

DisSModel's scientific lineage is rooted in the TerraME/LuccME research program at INPE. The submitting author conducted doctoral research at INPE under Prof. Gilberto Câmara and Dr. Ana Paula Dutra Aguiar — principal architects of TerraME/LuccME — and has co-authored the modeling program since 2009 (Costa et al., 2009; Moreira et al., 2009); the DisSLUCC packages reimplement in Python the continuous and discrete allocation components of that lineage (Aguiar et al., n.d.). On 7 May 2026, DisSModel was presented at INPE's Graduate Program in Applied Computing seminar series (recording: <https://youtu.be/o7pMJt0CvXU>), connecting the framework to the institutional community that maintains TerraME and LuccME.

The framework is in active use across two UFMA research groups. Within LambdaGeo, graduate students develop disslucc-continuous and brmangue-dissmodel in their Master's

research. Independently, Prof. Denilson da Silva Bezerra (UFMA, former INPE), whose doctoral work established BR-MANGUE's scientific foundation (D. da S. Bezerra et al., 2013), uses the DisSModel reimplementation in his own coastal dynamics program (PVCBS4959-2025, PVCBS4960-2025; <https://sigaa.ufma.br/sigaa/public/docente/pesquisa.jsf?siape=3104707>), a collaboration predating DisSModel itself (D. da S. Bezerra et al., 2025).

Starting August 2026, the project receives its first undergraduate research fellows, funded by UFMA and by CNPq, one of them supervised by a collaborating faculty member. The 2026 development effort was oriented toward this milestone: stabilizing the ModelExecutor contract so each fellow can own an independent repository — `dislucc-continuous`, `dislucc-discrete`, `brmangue-dissmodel`, or `disscube` (a data-cube layer, the Python successor to TerraME's `fillCellularSpace`) — without core changes.

Since the original submission, development has continued with `dislucc-discrete` (Costa, 2026), a CLUE-S-like discrete allocation package using logistic regression — the discrete counterpart to DisSLUCC-Continuous. An initial version has been validated against the Lab15 case study (Moju municipality, 5,914 cells, 6 steps) from the reference LuccME implementation (Aguar et al., n.d.), reaching cell-for-cell agreement — zero quantity and zero allocation disagreement (Pontius Jr. & Millones, 2011) — at 56.8 ms/step. A shipped discriminance test shows this scenario is also reproduced by a trivial static ranking, so it validates coefficient transcription rather than the allocation algorithm; a dynamic-covariate scenario is planned.

These packages — `dissmodel-ca`, `dissmodel-sysdyn`, `DisSLUCC-Continuous`, `dislucc-discrete`, and `brmangue-dissmodel` — demonstrate that the ModelExecutor contract generalizes across modeling paradigms without core modifications. Studies such as F. G. S. Bezerra et al. (2022), developed using LuccME, are the class of models the DisSLUCC packages aim to reproduce. A roadmap toward DisSModel 1.0 (May 2027) anchors community outreach including an open textbook, *Geospatial Modeling with Python* (<https://lambdageo.github.io/geospatial-modeling-python/>), already in progress. This positions DisSModel as the simulation layer in the Brazilian Earth Observation stack, complementary to SITS (Simoes et al., 2021) and the Brazil Data Cube (Ferreira et al., 2020).

Author Contributions

CRediT roles: **S.S.C.** — Conceptualization, Software (Core, Geo, Executor, IO, Visualization), Methodology, Validation, Writing, Supervision, Project administration. **N.J.S.J.** — Conceptualization, Software (initial design), Validation, Writing (undergraduate thesis (Santos Junior, 2025)). **D.S.B.** — Conceptualization (domain science), Validation, Resources (D. da S. Bezerra et al., 2013, 2025). **F.M.S.** — Software (`brmangue-dissmodel`), Data curation, Validation (D. da S. Bezerra et al., 2025). **J.M.P.A.** — Software (`dislucc-continuous`), Validation. All authors reviewed and approved the final manuscript.

AI Usage Disclosure

Development followed several phases. The initial prototype (May–June 2025), corresponding to the second author's undergraduate thesis (Santos Junior, 2025), did not involve generative AI. Development resumed in February–March 2026 with Claude (chat) used mainly for documentation; from April 2026, Gemini CLI accelerated code generation and refactoring; from June 2026, Claude Code (CLI) was used on newer satellite repositories such as `dislucc-discrete`, including the audit of the validation routines against the original TerraME scripts. AI tools also assisted with writing in English, not the submitting author's native language. The scientific design — the TerraME compatibility contract, executor pattern, dual-substrate architecture, and validation methodology — predates and is independent of this AI-assisted phase, tracing to the submitting author's doctoral research at INPE and the undergraduate thesis cited above. AI assisted with implementation velocity and language clarity, not scientific or architectural decisions. All outputs were reviewed and validated by the authors.

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